

From ML Signals to Investable Strategies

An AI Supply-Chain Pilot Study

BMF5360 Machine Learning in Investments

Midterm Report

Group #

[Member name, NUS e-number] · [Member name, NUS e-number] · [Member name, NUS e-number]

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1. Project idea

This project builds a machine-learning investment research framework and tests it on a first pilot universe, US-listed companies in the AI supply chain. The framework fixes six steps before any result is read. These are a universe with dated membership, features with a fixed information boundary, a supervised target, a training/validation/sealed-test split, a pre-registered mapping from predictions to positions, and a written rule for choosing the final fund strategy. The AI pilot asks a narrow question inside this framework. Does information available at the close of trading session F-1 help identify which AI supply-chain stocks will beat SPY over the following 21 trading sessions? The pilot is a completed prototype, not a claim that the final fund must be an AI-only fund. The final strategy will be chosen at the next stage with the criteria in Section 5.

1.1 How the idea was selected

We screened ten candidate strategies against LSEG Workspace data access on 7 September 2026, including regime-driven asset allocation, post-earnings-announcement drift, news-price fusion, FX carry, and the volatility risk premium. We first pursued earnings-information diffusion on the full S&P 500 universe, in which the earnings surprise of one company is used to predict the returns of economically linked companies. A gradient-boosting model on a shared-analyst-coverage graph produced a mean event-level Spearman correlation of -0.0040 in validation, against 0.0199 for a ridge control model. The paired difference was -0.0239 with a 95% block-bootstrap interval of [-0.0523, 0.0081]. We recorded this as a negative result and did not revise it with later subsamples.

We then moved to the AI supply chain for two reasons. The economic transmission channel is explicit, from accelerators to memory, servers and networking, cloud platforms, and data-centre power, so the hypothesis that information travels along the chain can be tested with a defined graph. Membership discipline is also the central methodological problem for any thematic strategy, which makes it a good place to test the framework. A static list of today's AI names would use 2026 knowledge to choose 2015 holdings. PLTR listed in September 2020, VRT listed through a SPAC in 2020, and NVIDIA in 2015 earned most of its revenue from gaming GPUs. The pilot therefore starts from a point-in-time S&P 500 parent universe and records the membership evidence for each company.

1.2 Why machine learning

The candidate signals are many and individually weak. The pilot uses 62 variables covering price technicals, liquidity, market state, macro state, and AI-industry state. Predicting whether a stock's 21-session return exceeds SPY's is a supervised binary classification problem, and a classifier can weigh these variables jointly under a fixed protocol. Logistic Regression is the transparent benchmark because its coefficients can be reviewed and it tests whether a more flexible model adds anything. Random Forest is the matched non-linear comparison. The ML contribution is measured as a pre-specified improvement over the simpler model with dependence-aware uncertainty, never as the highest metric found by searching the validation set.

2. Data

2.1 Sources and parent universe

Market data come from LSEG Workspace. We rebuilt a point-in-time S&P 500 universe of 781 RICs, including 154 delisted identifiers, by rolling the 7 September 2026 constituent list backwards through 751 index-change records. The cleaned daily panel holds about 2.08 million company-day rows of adjusted prices, volume, and bid/ask quotes from 2015 to 2026, plus 28,995 earnings-announcement

events with announcement timestamps. Business-segment revenue and R&D with original announcement dates were collected for the AI-exposure work. Macro data come from FRED (VIX, 3-month, 2-year, and 10-year Treasury yields, the broad dollar index, and high-yield and investment-grade OAS; 7 series, 18,345 observations). ETF prices for SOXX, XLK, AIQ, BOTZ, and IGV describe the AI-industry state.

2.2 Pilot universe

The pilot pool holds 49 RICs in seven supply-chain groups (Table 1). Companies were selected from the 781-RIC parent universe by economic role, using segment disclosures and company descriptions. Historical returns, volatility, size, liquidity, and model output played no part in selection. Delisting is not an exclusion criterion; JNPR.N[^]G25 stays in the registry with its membership ending on 8 July 2025. Evidence quality is recorded separately from economic role. Nine companies have direct local point-in-time evidence, meaning historical segment disclosures joined to original announcement dates. The other 40 have provisional static evidence from current company descriptions, which cannot prove membership in earlier years. We therefore call the 49 an exploratory pool and treat strict historical membership as next-stage work.

Table 1. Pilot universe by supply-chain group

Group	Economic role	Companies	Direct PIT	Provisional
GPU / accelerator	GPUs, CPUs, custom accelerators	5	2	3
AI semiconductor	AI SoCs, analog, power and connectivity silicon	6	0	6
Memory / HBM / storage	HBM, DRAM, NAND, enterprise storage	5	1	4
Server / network	AI servers, switches, optical interconnect	6	1	5
Cloud / software	Cloud platforms, AI software, EDA	10	1	9
Data-centre power / cooling	Data centres, power, cooling, infrastructure	9	3	6
Robotics / autonomy	Industrial automation, vision, autonomous driving	8	1	7
Total		49	9	40

2.3 Features and information boundary

Every feature uses data available through session F-1, one session before the formation date F. The 21 technical, liquidity, and market variables are stock momentum over 1, 5, 20, and 60 sessions; 20- and 60-session volatility; 126-session beta, idiosyncratic volatility, and beta observation count; 20- and 60-session medians of volume, dollar volume, and quoted spread; and SPY momentum and volatility over the same windows. The 20 macro variables are the VIX level, its 1-, 5-, and 20-day changes and 252-day percentile; 3-month, 2-year, and 10-year yield levels with 5- and 20-day changes; the broad dollar level with 5-, 20-, and 60-day changes; and the 10y-2y and 10y-3m term spreads. The 21 AI-industry-state variables are SOXX and XLK momentum, volatility, and 60-day drawdown; SOXX return relative to SPY over 5, 20, and 60 sessions; and pool breadth, meaning the share of pool stocks rising or above their 20- and 60-day moving averages, cross-sectional return dispersion, and the equal-weight pool return and its excess over SPY. Six credit-spread variables were excluded because the local download starts on 11 September 2023, leaving no training or validation coverage. Eighteen rolling variables on AIQ, BOTZ, and IGV were excluded because those ETFs did not exist in the early training years and their missingness would act as a calendar label.

2.4 Target, split, and leakage controls

The target is $y = 1$ when the stock's 21-session total return from the close of F exceeds SPY's 21-session total return, and $y = 0$ otherwise. Training covers 2015-2020 with 2,334 non-overlapping 21-

session anchors across 39 companies and 444 formation dates. Validation covers 2021-2022 with 992 anchors, 44 companies, and 89 formation dates; 50.7% of validation labels are positive. The test period, 1 January 2023 to 30 June 2026, is sealed. Its labels have not been opened, hashed, predicted, or scored, and the code package does not list the sealed file as an input. Samples are never shuffled. Each company keeps at most one anchor per 21-session block, so labels do not overlap within a company. Random Forest tuning uses expanding time folds ending in 2018, 2019, and 2020 within the training period, and every training row must exit before the fold's validation start, which gives a full 21-session purge.

2.5 Processing rules

Every action that changes data was listed, approved, and logged before it ran. Empty alignment cells created by the wide-format price download (167,134) and vendor padding rows after a delisting were quarantined with reason codes rather than deleted. Different RICs sharing a root, such as the pre-2019 and post-2019 Dow entities, are kept as separate securities. Pre-market announcements enter at the same-day close and after-market announcements at the next close. Daily returns beyond 20% in absolute value go to a review list and are neither removed nor winsorized. Missing values stay missing in every source table. The only imputation is inside the model pipeline, a median fitted on training rows with a missing indicator. There is no zero fill, forward fill, interpolation, or winsorization anywhere in the data layer.

3. Baseline models

Both models run as fixed scikit-learn pipelines on the same anchors. The Logistic pipeline is SimpleImputer (median, add_indicator), StandardScaler, and LogisticRegression with $C = 1.0$, class_weight = balanced, the lbfgs solver, max_iter = 2000, and random_state = 5360. The imputer and scaler are fitted on training rows only. The Random Forest uses 200 trees and an 18-point grid over max_depth, min_samples_leaf, and max_features; each feature set selects its own grid point by mean ROC-AUC across the three purged training folds, with Brier score as the tie-breaker. Four feature sets were registered before any validation score was read, namely technical only (21), technical plus macro (41), technical plus AI state (42), and the full set (62). The validation period was used once, for the final comparison of these eight models. We report ROC-AUC, PR-AUC, Brier score, accuracy at a 0.50 threshold, and the rank information coefficient (IC) between predicted probability and realised 21-session excess return, pooled and averaged by date.

4. Preliminary results

4.1 Classification comparison

Table 2. Validation results, 2021-2022, 992 anchors (same sample for every row)

Feature set	Model	ROC-AUC	PR-AUC	Brier	Accuracy	Pooled rank IC	Mean daily rank IC
Technical	Logistic	0.5395	0.5421	0.2511	51.61%	0.140	0.078
Technical	Random Forest	0.5231	0.5511	0.2543	51.92%	0.074	0.063
Technical + macro	Logistic	0.5616	0.5524	0.2508	56.45%	0.174	0.086
Technical + macro	Random Forest	0.5362	0.5454	0.2496	52.92%	0.096	0.036
Technical + AI state	Logistic	0.5799	0.5745	0.2476	54.54%	0.187	0.089
Technical + AI state	Random Forest	0.5391	0.5422	0.2520	54.84%	0.104	0.087
Full set	Logistic	0.5820	0.5456	0.2700	55.65%	0.186	0.093
Full set	Random Forest	0.5349	0.5481	0.2506	52.62%	0.088	0.037

The technical-plus-AI-state Logistic model is the current primary model. Its ROC-AUC of 0.5799 is 0.0405 above the technical-only Logistic benchmark, and a paired bootstrap clustered by formation date (2,000 draws) gives a 95% interval of [0.0007, 0.0801] for that lift. The model's AUC is 0.5906 in 2021 and 0.6213 in 2022, so the direction does not reverse across years. The full-set Logistic model has a marginally higher AUC (0.5820) but a Brier score of 0.2700 against 0.2476, so its probabilities are less reliable and it is not used for decisions. Random Forest is below Logistic in every feature set, and adding macro variables makes its diagnostic top-minus-bottom spread negative. Complexity did not add value in this pilot.

4.2 Timing versus selection

Pooled AUC mixes two different claims. Of the 41 technical-plus-macro variables, 27 are constant within a formation date (all 20 macro variables and 7 SPY-state variables), so part of the pooled score comes from ranking dates, not from ranking stocks. We therefore decomposed the macro contribution with a validation-only audit. The pooled AUC lift from macro variables is 0.0221 with a 21-session block-bootstrap interval of [-0.0394, 0.0704]. Restricted to pairs of stocks on the same formation date, the pair-weighted within-date AUC lift is 0.0062 with an interval of [-0.0066, 0.0218], and the within-date rank-IC lift is 0.0082 with an interval of [-0.0074, 0.0240]. Every interval crosses zero. We therefore do not claim that macro variables improve stock selection, and we read the pooled AUC as a mixed timing and selection diagnostic.

4.3 Validation-period portfolio

To test whether the classifier's ranking converts into returns, we pre-registered a portfolio rule before running it and did not change it afterwards. Every 21 sessions the primary model scores all eligible pool stocks; stocks with `p_up` at or above 0.55 are held with a 10% single-stock cap and a 30% supply-chain-group cap; a date with fewer than five qualifying stocks is skipped; the sleeve is scaled to a 10% volatility target, capped at 50% one-way turnover per rebalance, hedged with SPY at the estimated beta, and charged half the quoted spread on every trade. Table 3 reports the result over 2021-2022 (503 scored sessions, 23 holding blocks, 4 blocks skipped).

Table 3. Pre-registered validation portfolio, 2021-2022, net of spread costs

Metric	Value	Metric	Value
Cumulative net return	-3.10%	Beta to SPY	0.064
Annualised return	-1.63%	Annualised alpha (cash = 0)	-1.69%
Annualised volatility	9.78%	One-way turnover, annualised	4.54x
Sharpe ratio (cash = 0)	-0.119	Spread cost / initial NAV	0.24%
Maximum drawdown	-13.15%	SPY cumulative return, same period	+13.11%

The portfolio lost money while the classifier's validation metrics improved, and the 0.24% spread cost does not explain the loss. An independent daily re-accounting of cash, positions, trades, and costs passed 13 of 14 checks with one non-blocking warning. We keep this negative result as the honest validation outcome. Re-tuning the 0.55 threshold or the caps on validation data would turn the validation period into a second training set, so the next step is attribution on the frozen rule, not a search for a better-looking number.

5. Limitations and next stage

The pilot has established that a 49-company AI supply-chain universe can be documented by role, membership span, and evidence source; that a daily 21-session prediction pipeline can keep a training-only preprocessing rule and a sealed test period; and that Logistic Regression is a valid benchmark that outperformed the matched Random Forest. It has not established a tradable signal. Four

limitations govern how the numbers should be read. First, 40 of 49 companies lack point-in-time membership evidence, so survivorship bias remains possible. Second, the FRED download is not a vintage database, so macro values may include later revisions. Third, the 992 validation rows share market shocks across stocks on the same date, so they are not independent observations, and all intervals above use date or block resampling. Fourth, probabilities are not yet calibrated; the expected calibration error is 0.0573 overall and rises to 0.1247 in 2022, which is why the portfolio rule uses cross-sectional ranks and a fixed threshold rather than probability levels.

The next stage has five steps. We will attribute the negative validation return to the long sleeve, the SPY hedge, the skipped dates, and the group and single-name contributions. We will then pre-register a second portfolio rule (rebalance on a fixed session each month, hold the top 20% by score at equal weight, and report one-way costs of 0, 5, 10, and 25 basis points), choosing its settings on training data only. We will build strict historical membership for the 40 provisional names or restrict the universe to the nine with direct evidence. We will then choose the final fund strategy with the criteria in Table 4, and only after every design choice is frozen will we open the sealed test period once and report out-of-sample Sharpe, drawdown, turnover, and net returns.

Table 4. Criteria for selecting the final fund strategy

Criterion	Evidence required before selection
Economic mechanism	A falsifiable reason why returns or risk should differ from the benchmark
Point-in-time feasibility	Dated universe membership and feature availability, including delistings
Sample adequacy	Enough independent dates or events after accounting for holding-period overlap
ML contribution	A pre-specified improvement over a simple model or rule, with dependence-aware uncertainty
Implementation	A frozen portfolio rule with costs, turnover, capacity, and risk controls
Test discipline	One untouched final period opened once after all choices are fixed

Reproducibility. All figures come from frozen, hashed runs. The code package reruns the technical and technical-plus-macro baselines from the frozen model-ready inputs and never reads the sealed test file. Data processing, exclusions, and audit checks are recorded in DATA_PROCESSING_LOG.md, and material AI assistance is recorded in AI_USE_LOG.md.

AI tool declaration. We used OpenAI Codex and Claude Code in preparing this project to draft data-collection and modelling scripts, run audits, and improve the clarity of this report. All research design decisions, data-processing approvals, and interpretations are our own, and we are responsible for the content and quality of the submitted work.

References

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